4 **(a)** **(i)** State the principle of superposition.

.....

.....

..... [2]

(ii) Use Huygen's principle to explain the interference pattern formed when waves of a single wavelength pass through a single slit.

.....

..... [1]

(b) A parallel beam of light of wavelength 600 nm from a point source is incident normally on a rectangular slit of width 0.30 mm as shown in Fig. 4.1. Light passing through the slit is incident on a screen placed a distance *D* from the slit.

The centre of the interference pattern formed on the screen is at O. The angle a light ray emerging from the slit makes with the normal line between the slit and the screen is θ.

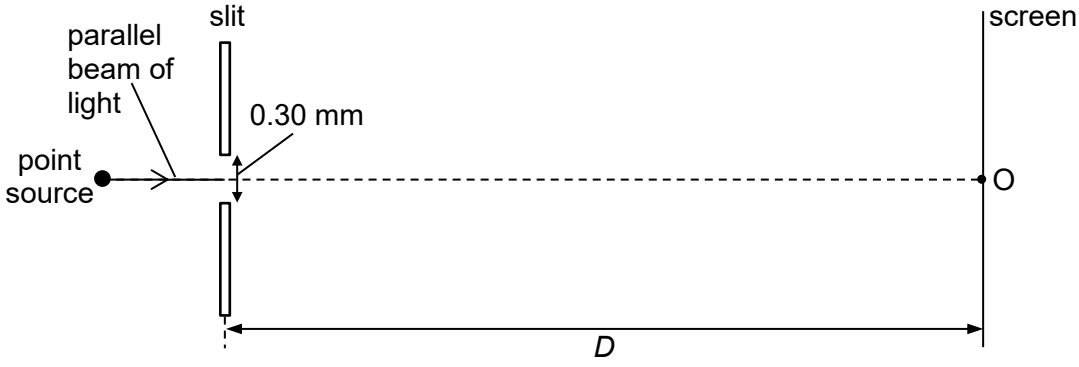

Fig. 4.1 (not to scale)

(i) On the axes in Fig. 4.2, sketch a graph to show the variation with $\sin\theta$ of the intensity I of the light on the screen. Include appropriate values along the $\sin\theta$ axis.

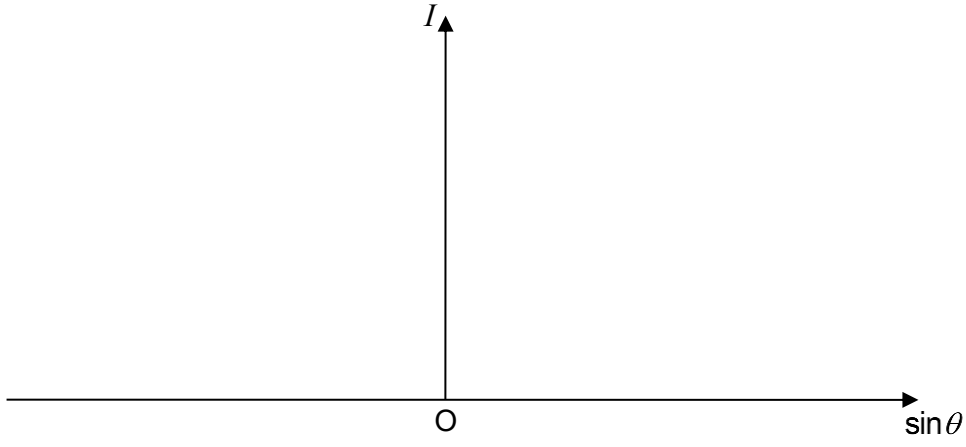

Fig. 4.2

[3]

(ii) Another identical point source of light is placed at a distance of 1.0 mm from the first point source as shown in Fig. 4.3. Both sources are 0.25 m from the slit. Light from each source forms a separate interference pattern on the screen.

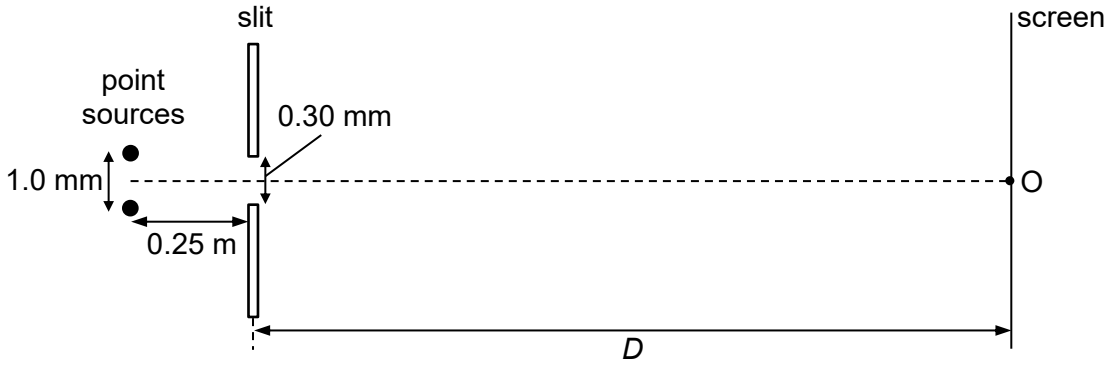

Fig. 4.3 (not to scale)

Determine, with appropriate calculations, if it is possible to resolve the images of the two sources on the screen.

The two interference patterns can / cannot be resolved. [2]

(c) The single slit in Fig. 4.1 is replaced with double slits each with the same slit width of 0.30 mm and with a slit separation of 1.2 mm. There is only one point source and light from the point source is incident normally on the slits.

(i) Determine the number of maxima observed within the central fringe of the single slit interference pattern. Explain your working clearly.

number of maxima = ... [3]

(ii) If the width of the central fringe is 12 mm, calculate the perpendicular distance D between the double slits and the screen.

D = ... m [2]